IBOOKTRACK · HANDOVER DOCUMENTATION

Backend Documentation

A unified Warehouse Management System and Executive Dashboard for Innodata's book-digitization pipeline.

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iBookTrack — Backend Documentation

doc backend Node.js 20+ Express 4 PostgreSQL

Express.js API served from `backend/`. Runs on port **5000** by default.

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Tech Stack

Package	Version	Purpose
`express`	^4.21	HTTP server
`pg`	^8.11	PostgreSQL client pool
`jsonwebtoken`	^9.0	Custom JWT Auth
`cors`	^2.8	CORS middleware
`express-rate-limit`	^8.3	Login / register rate limiting
`dotenv`	^16.4	Loads `../.env`
`axios`	^1.14	HTTP for external APIs
`mssql`	^12.2	(legacy direct-DB integration, optional)
`xlsx`	^0.18	Excel parsing for `scanning-excel-reader`

ES modules everywhere (`"type": "module"` in `package.json`).

Scripts:

- `npm run dev` — `node --watch server.js` (auto-restart on save)
- `npm start` — `node server.js`

Directory Layout

```
backend/
├── server.js                    # entry point + CORS + route registration
├── config/
│   └── db.js                    # exports { pool }
├── middleware/
│   └── authMiddleware.js        # requireAuth (Bearer token validator)
├── routes/                      # one file per resource
│   ├── auth.js                  # register / login / logout / refresh / me
│   ├── boxes.js                 # CRUD + advance/rollback/receive
│   ├── books.js                 # ISBN search + flag/unflag
│   ├── lots.js                  # Lot CRUD
│   ├── warehouse.js             # Sessions + scans + rejected-books
│   ├── dashboard.js             # Aggregates for the home dashboard
│   ├── reports.js               # Reports / exports
│   ├── scanning.js              # Read-only Simflex stats
│   ├── storage.js               # Storage locations
│   └── index.js                 # ⚠️  unused route
├── services/
│   ├── ilms-api.js              # Innodata iLMS HTTP client
│   ├── simflex-api.js           # Simflex HTTP wrapper
│   ├── simflex-db.js            # Higher-level Simflex queries
│   ├── simflex-poller.js        # Polls Simflex every 60s
│   └── scanning-excel-reader.js # Legacy XLSX importer
├── migrations/                  # 25 numbered SQL files + README.md
├── utils/
│   └── testConnection.js        # logs DB connectivity at boot
└── test-warehouse.js            # manual smoke test (delete pre-prod)
```

Boot Sequence

`server.js` :

1. Load `../.env`
2. Build CORS allowlist from `FRONTEND_URL` / `FRONTEND_URLS` + private LAN
3. `app.use(express.json())`
4. Register routers under `/api/<resource>`
5. Health check: `GET /api/health` → `{ status: 'ok', timestamp }`
6. `app.listen(PORT)` then:
   - `testConnection()` — pings PostgreSQL
   - `startSimflexPoller()` — kicks off the 60s polling loop

Authentication & Middleware

Database client (`config/db.js`)

- `pool` — `pg` connection pool used for all database queries.

`requireAuth` (`middleware/authMiddleware.js`)

Reads `Authorization: Bearer <jwt>`, validates with `jsonwebtoken`, and attaches `req.user` (decoded JWT payload).

Returns `401` on missing/invalid token. The frontend's `authFetch` automatically redirects to `/login` on 401.

API Routes

All routes are JSON in / JSON out. Errors return `{ error: string }` with an appropriate 4xx/5xx code.

`/api/auth` (`routes/auth.js`)

Method	Path	Auth	Body	Notes
POST	`/register`	-	`{ name, email, password }`	Rate-limited 5/hr/IP. Creates `profiles` row.
POST	`/login`	-	`{ email, password }`	Rate-limited 10/15min/IP. Returns tokens.
POST	`/refresh`	-	`{ refresh_token }`	Returns new `access_token` + `expires_at`.
POST	`/logout`	✅	-	Revokes session.
GET	`/me`	✅	-	Echoes profile + user.

`/api/boxes` (`routes/boxes.js`)

Method	Path	Auth	Description
GET	`/`	✅	List boxes. Filters: `lot_id`, `current_stage`, `box_number`. Includes scan count.
GET	`/:id`	✅	Single box + all `warehouse_book_scans` joined.
GET	`/lot/:lot_number`	-	Boxes for a lot.
GET	`/stats/pipeline`	✅	Counts per `current_stage` (optional `lot_number` filter).
GET	`/pipeline-transitions`	✅	Recent `pipeline_events` (limit param).
POST	`/`	✅	Create box.
PUT	`/:id`	✅	Update stage / counts. Auto-logs `pipeline_events` on stage change.
POST	`/:id/receive`	✅	Mark received at the current stage (sets `received_at`).
POST	`/:id/advance`	✅	Advance to next stage (`STAGE_ORDER` list). Logs event.
POST	`/:id/rollback`	✅	Body `{ to_stage }`. Reverts and logs.

`STAGE_ORDER = ['warehouse','inventory','cut','endorsement','scan','transmit','storage']`

`/api/books` (`routes/books.js`)

Method	Path	Auth	Description
GET	`/`	✅	List books with filters.
GET	`/isbn/:isbn`	✅	Lookup by ISBN.
GET	`/ilms/search/:isbn`	-	Proxy to iLMS for live metadata.
GET	`/flags`	✅	Bulk-fetch flags. Query: `scan_ids=id1,id2,…`
POST	`/`	✅	Create book record.
POST	`/:id/flag`	✅	Body `{ issue, remarks, box_id }`. Records QA flag.
PUT	`/:id`	✅	Update.
POST	`/:id/transmit`	✅	Mark transmitted.
POST	`/:id/unaccounted`	✅	Mark missing.
POST	`/scan`	✅	Inventory-gate scan log.

`/api/lots` (`routes/lots.js`)

CRUD for `lots` plus `GET /:lot_number` lookup.

`/api/warehouse` (`routes/warehouse.js`)

Method	Path	Description
POST	`/sessions/start`	Begin a receiving session for a lot.
POST	`/sessions/:session_id/scan`	Save one scan. Body: `{ isbn, scan_status, rejection_reason?, notes? }`. Auto-creates a box on first scan.
GET	`/sessions/:session_id`	Session + all scans.
POST	`/sessions/:session_id/complete`	Close session.
GET	`/sessions`	List sessions.
GET	`/search/isbn/:isbn`	Find which box holds an ISBN.
GET	`/rejected-books`	Non-processable scans (`limit` param). Joins box → lot.
PATCH	`/scans/:scan_id/flag`	Re-flag an existing scan.
POST	`/backfill-titles`	Maintenance: refill `title` field via iLMS.

`scan_status` ∈ `{ 'processable', 'non_processable' }`. `rejection_reason` is one of the 23 canonical flag keys (see Project Overview).

`/api/dashboard` (`routes/dashboard.js`)

Aggregate endpoints powering the home dashboard cards/charts: `/stats`, `/transmission-queue`, `/unaccounted-books`, `/pipeline-stats`, `/scan-history`. All accept optional `lot_number` filtering.

`/api/scanning` (`routes/scanning.js`)

Read-only proxy over Simflex. See Services.

| GET | `/summary` | Today's totals. | | GET | `/by-machine` | Per-machine breakdown. | | GET | `/by-stage` | Stage breakdown. | | GET | `/data` | Range query (`startDate`, `endDate`). | | GET | `/live` | Combined snapshot. |

`/api/reports` (`routes/reports.js`)

Composable report generators backing the Reports page.

`/api/storage` (`routes/storage.js`)

Locations CRUD for the warehouse locator.

Services (External Integrations)

Simflex WMS (`services/simflex-*.js`)

- **Transport:** HTTPS to `https://simflex.innodata.com/wms.services/WMS.Services.asmx/TechGetDatabaseDataset`
- **Auth:** API key + config code passed as query params.
- **Config:** read from `.env` (`SIMFLEX_API_URL` , `SIMFLEX_API_KEY` , `SIMFLEX_CONFIG`). The poller logs a warning if any are missing.
- **Polling:** `simflex-poller.js` runs every 60s on boot, caches the latest summary used by `/api/scanning/live` and the dashboard live tiles.

iLMS (`services/ilms-api.js`)

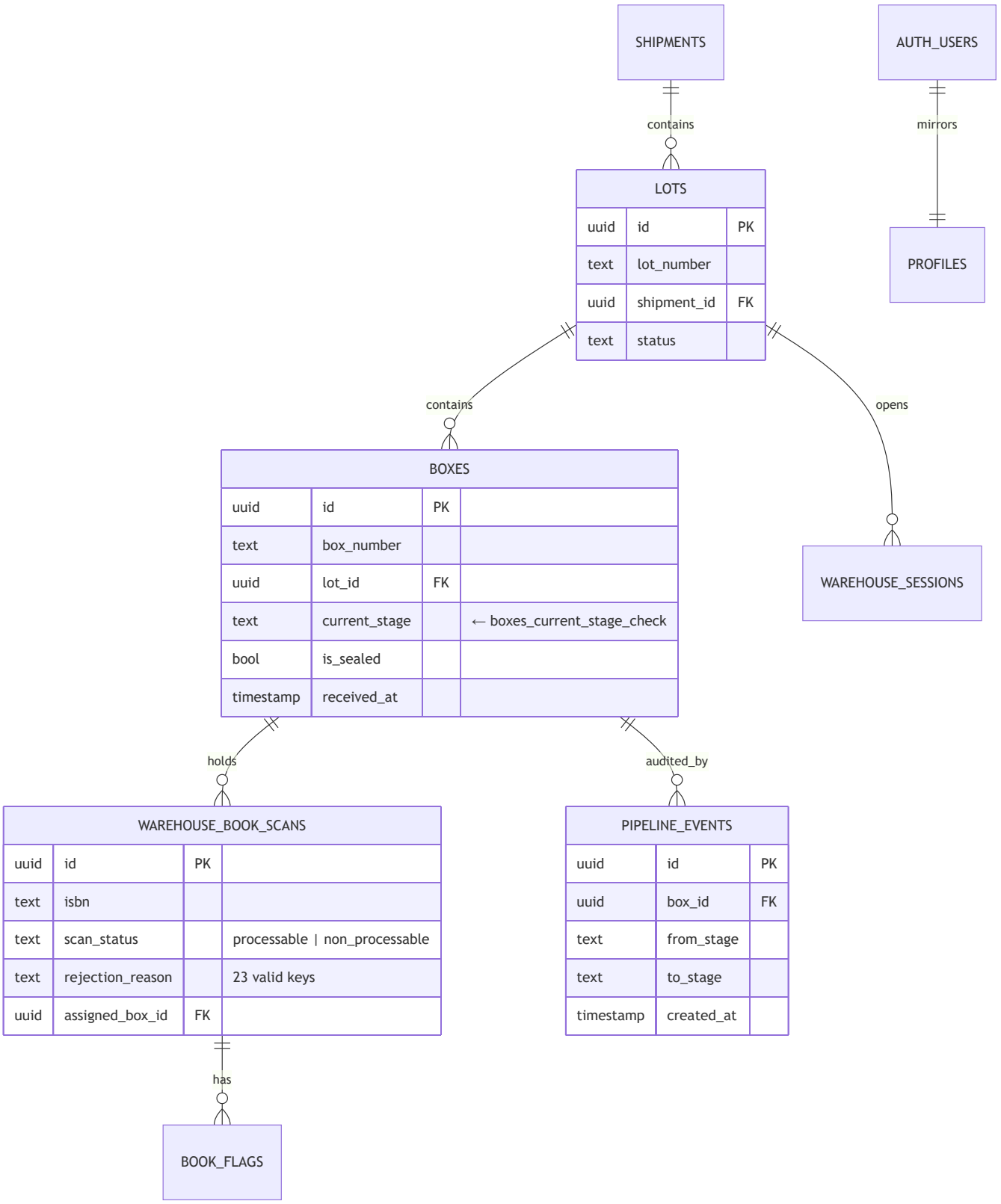
Looks up book metadata by ISBN. Used at warehouse-inventory scan time so the scanner sees the title immediately.

Excel reader (`services/scanning-excel-reader.js`)

Legacy bulk import of scan logs from XLSX. Kept for one-off recoveries; safe to remove once you confirm nobody runs it.

Database Schema

Hierarchy



Cross-cutting tables:

- `profiles` — mirrors `auth.users`, holds `name`, `email`, `role`, `profile_picture` URL.
- `pipeline_events` — append-only audit log for every stage change.
- `warehouse_sessions` — open scanning sessions; FK → `lots`, `boxes`.
- `transmission_queue` — progress rows surfaced on the dashboard.
- `unaccounted_books` — flagged-as-missing rows per stage.
- `scan_history` — inventory-gate scan log.
- `storage_locations` — the locator grid.
- `staging_*` — temporary import targets (Indic comparisons, etc.).

Detailed column lists are in each migration file under `backend/migrations/`.

Database Constraints

These are the "land mines" we hit during dev. Document them when you change stages or flag values.

`boxes_current_stage_check`

Allowed values:

```
warehouse, inventory, cut, endorsement, scan, transmit, storage, intake
```

> **CAUTION**
>
> **To change this constraint, ALWAYS go in this exact order:**
>
> 1. `ALTER TABLE boxes DROP CONSTRAINT boxes_current_stage_check;`
> 2. `UPDATE boxes SET current_stage = '<new>' WHERE current_stage IN ('<legacy>', …);`
> 3. `ALTER TABLE boxes ADD CONSTRAINT boxes_current_stage_check CHECK (current_stage IN (…));`
>
> Doing the ADD before the UPDATE will fail because existing rows still hold the old value. This has bricked dev DBs more than once during turnover.

`warehouse_book_scans.scan_status`

Allowed values: `processable`, `non_processable`. (Old data may contain `accepted`, `rejected` — migrated in early 2026.)

`warehouse_book_scans.rejection_reason`

23 valid keys (21 modern + 2 legacy). Source of truth lives in the frontend `FLAG_ISSUES` array; keep both in sync. Listed in `PROJECT_OVERVIEW.md`.

Row-Level Security

Every public table has RLS enabled. Anonymous and authenticated users have read-only or row-scoped access; admins via service-role bypass everything. , `011` , `013` , `019` .

Migrations

Files `001` through `025` (note `020` is intentionally absent — numbering gap, not a missing migration). Run in order on a fresh DB.

Critical ones:

- `005_create_boxes_table.sql` — defines `current_stage` enum
- `012_create_warehouse_operations.sql` — sessions + scans
- `014_add_box_capacity_and_sealing.sql` — `is_sealed` , capacity per box
- `024_fix_pipeline_events_constraint.sql` — adds `endorsement` to allowed events
- `025_optimize_lot_stats.sql` — `get_lot_stats()` RPC used by dashboard

Background Jobs

- **Simflex poller** — `services/simflex-poller.js` , started in `server.js` . 60s interval. Caches the last successful payload and exposes it through `/api/scanning/live` .

There are no other cron jobs. Anything time-based should live here.
